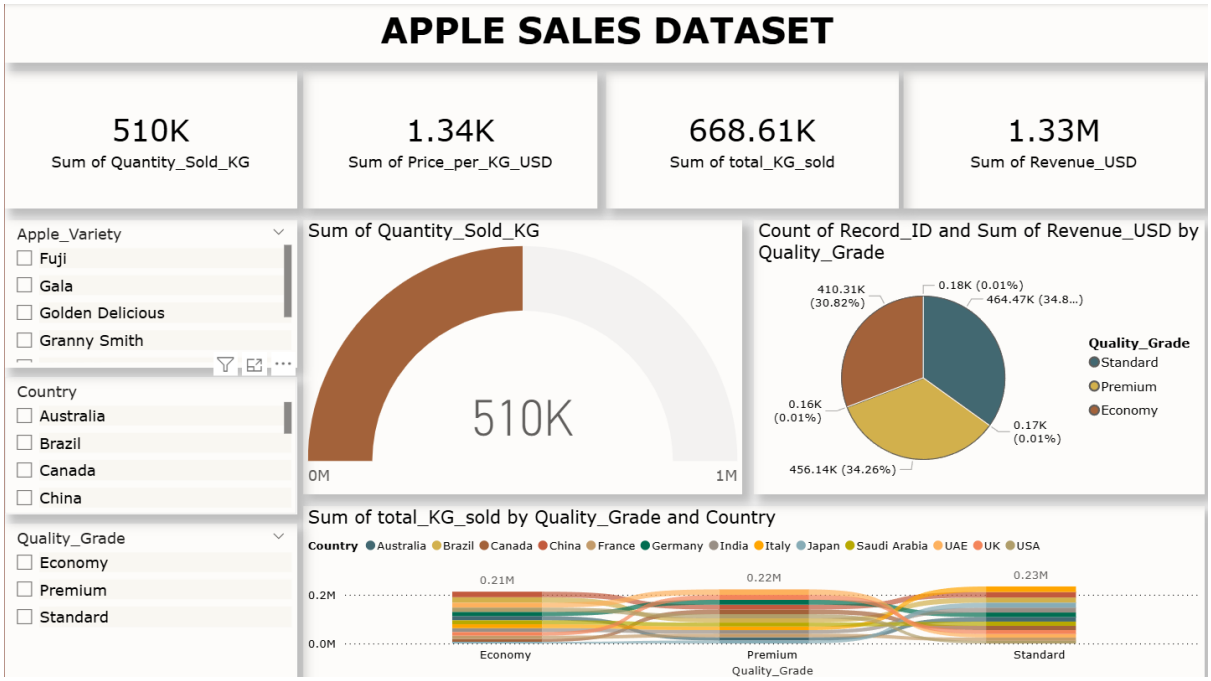
APPLE SALES DATA ANALYSIS DASHBOARD

INTRODUCTION

Data analysis helps organizations understand sales patterns, customer demand, and product performance. This project focuses on analyzing apple sales data using **data visualization techniques**. The dashboard was developed using **Microsoft Power BI** to convert raw sales data into meaningful insights.

The goal of this project is to analyze apple sales across different **countries, apple varieties, and quality grades** to identify trends and support better decision-making.



OBJECTIVES OF THE PROJECT

The main objectives of this project are:

- To analyze apple sales using a dataset.
- To visualize sales performance through an interactive dashboard.
- To identify top-performing apple varieties and quality grades.
- To compare apple sales across different countries.
- To provide insights that help improve sales strategies.

DATASET DESCRIPTION

The dataset used in this project contains information related to apple sales. The important columns in the dataset include:

Column Name	Description
Record_ID	Unique identifier for each record
Apple_Variety	Type of apple (Fuji, Gala, Golden Delicious, Granny Smith)
Country	Country where apples were sold
Quality_Grade	Category of apple quality (Economy, Premium, Standard)
Quantity_Sold_KG	Quantity of apples sold in kilograms
Price_per_KG_USD	Price of apples per kilogram
Total_KG_Sold	Total kilograms sold
Revenue_USD	Total revenue generated

The dataset helps analyze the relationship between **quantity sold, pricing, and revenue**.

TOOLS AND TECHNOLOGIES USED

Tool	Purpose
Microsoft Power BI	Data visualization and dashboard creation
Microsoft Excel	Data storage and preprocessing
Data Analytics	Extracting insights from the dataset

DATA PREPROCESSING

Before creating the dashboard, the dataset was cleaned and prepared.

Steps performed:

- Removed duplicate records
- Checked for missing values
- Verified correct data types
- Organized columns properly
- Loaded the dataset into **Microsoft Power BI**

DASHBOARD COMPONENTS

The Apple Sales Dashboard contains several visual elements.

1. KPI Cards

The dashboard displays key performance indicators such as:

- **Total Quantity Sold:** 510K KG
- **Price per KG:** 1.34K USD
- **Total KG Sold:** 668.61K
- **Total Revenue:** 1.33M USD

These metrics help quickly understand overall sales performance.

APPLE VARIETY FILTER

Users can filter data based on apple types such as:

- Fuji
- Gala
- Golden Delicious
- Granny Smith

COUNTRY FILTER

The dashboard allows filtering sales by country, including:

- Australia
- Brazil
- Canada
- China
- France
- Germany
- India
- Italy
- Japan
- Saudi Arabia
- UAE
- UK
- USA

This helps compare international apple sales.

QUANTITY SOLD GAUGE CHART

A **gauge chart** displays the total quantity sold, showing **510K KG** out of a maximum value.

This visual helps quickly identify how close the sales are to the target.

QUALITY GRADE PIE CHART

The pie chart shows revenue distribution across quality categories:

- Standard
- Premium
- Economy

This visualization helps understand which quality grade contributes the most revenue.

COUNTRY VS QUALITY GRADE VISUALIZATION

This chart shows **Total KG Sold by Country and Quality Grade**.

It helps identify:

- Which country sells the most apples
- Which quality grade is most popular in different countries

KEY INSIGHTS FROM THE DASHBOARD

From the analysis, several insights were discovered:

1. **Standard quality apples generate the highest revenue.**
2. **Premium apples also contribute significantly to overall sales.**
3. Apple sales are distributed across multiple countries, showing strong global demand.

4. Some apple varieties perform better than others depending on the country.
5. Total revenue from apple sales reached **1.33 million USD**.

ENEFITS OF THE DASHBOARD

The dashboard provides the following benefits:

- Easy visualization of complex sales data
- Quick identification of sales trends
- Better decision-making for apple suppliers
- Real-time filtering and analysis

CONCLUSION

This project successfully demonstrates how **data analytics and visualization** can be used to analyze sales data. Using **Microsoft Power BI**, raw apple sales data was transformed into an interactive dashboard.

The dashboard provides valuable insights about apple varieties, country-wise sales, and quality grades. These insights can help businesses improve their marketing and sales strategies.

FUTURE ENHANCEMENTS

Future improvements for this project include:

- Predictive sales analysis using machine learning
- Integration with real-time sales databases
- Adding more advanced visualizations
- Creating automated sales forecasting models